Class Example :

Question 1 - Queue Implementation with Array:

You are required to implement a queue data structure using an array in C. The queue should have the following functionalities:

a) void enqueue(int item): This function should enqueue an integer item into the queue.

b) int dequeue(): This function should remove and return the front item from the queue.

c) int front(): This function should return the front item from the queue without removing it.

d) int isEmpty(): This function should return 1 if the queue is empty, and 0 otherwise.

Write the C code for the queue implementation with an array and demonstrate its usage by enqueuing a few elements, dequeuing elements, and checking whether the queue is empty.

Question 2 - Queue Implementation with Linked List:

You are required to implement a queue data structure using a singly linked list in C. The queue should have the following functionalities:

a) void enqueue(int item): This function should enqueue an integer item into the queue.

b) int dequeue(): This function should remove and return the front item from the queue.

c) int front(): This function should return the front item from the queue without removing it.

d) int isEmpty(): This function should return 1 if the queue is empty, and 0 otherwise.

Write the C code for the queue implementation with a linked list and demonstrate its usage by enqueuing a few elements, dequeuing elements, and checking whether the queue is empty.

Please ensure that your implementations handle edge cases appropriately and provide the correct output for different scenarios.

Question 3 - Circular Queue Implementation with Array:

You are required to implement a circular queue data structure using an array in C. The circular queue should have the following functionalities:

a) void enqueue(int item): This function should enqueue an integer item into the circular queue.

b) int dequeue(): This function should remove and return the front item from the circular queue.

c) int front(): This function should return the front item from the circular queue without removing it.

d) int isEmpty(): This function should return 1 if the circular queue is empty, and 0 otherwise.

e) int isFull(): This function should return 1 if the circular queue is full, and 0 otherwise.

Write the C code for the circular queue implementation with an array and demonstrate its usage by enqueuing a few elements, dequeuing elements, checking whether the queue is empty, and verifying if the queue is full.

Ensure that your implementation handles the circular nature of the queue correctly and provides the correct output for various scenarios.

Question 4 - Implement a Queue using Stacks:

You are tasked with designing a queue data structure using two stacks. Each stack should have push, pop, and isEmpty functions. The queue should support the following operations:

enqueue(int x): Add element x to the back of the queue.
dequeue(): Remove and retrieve the element from the front of the queue.

Dequeue constant : https://www.geeksforgeeks.org/queue-using-stacks/
Enqueue constant : Code uploaded